Impact of COVID on Campaign Measurement

8/11/20

COVID has led to significant and continual changes in consumer behavior which is expected to cause paradigm shifts in what we will call “the new normal”. With a possible second wave (or some might say a continuation of the first wave), and without a coordinated national response, we expect to see continued **differential consumer behavior across states and DMAs**, until there is a readily available vaccine or effective treatment.

With these continued disruptions to our society, there will be **irregularities in observed media performance**. Therefore, it becomes imperative to analyze COVID’s impact on campaign measurement, the adjustments being made to safeguard against these disruptive changes and how we are determining campaign success during COVID.

Key Considerations for PepsiCo

Results

What is the impact of COVID on campaign results?

Measurement Tools

What, if any, impact is there on our campaign measurement tools?

Methodology

What adjustments in methodology or interpretation should we make (if any)?

Performance Evaluation

How should we evaluate results for campaigns occurring during the COVID period?

Topline Findings

1. With some modifications, most of PepsiCo’s primary measurement tools should remain **accurate** in estimating ad performance during the COVID period
2. However, reported advertising **performance** should be viewed with caution as it reflects a phase of atypical consumer behavior and is likely to be irregular
3. Therefore, 2020 results will **not be directly comparable** to past periods and not projectable moving forward; channel results should be comparable in a relative sense but takeaways should be corroborated with historical findings

Recommendations Moving Forward

Measurement tools, recommended adjustments and POV on how we should utilize each

Measurement Solution	Recommended Adjustment	Solution Accuracy	Advertising Performance	Recommended Use/ Interpretation
MMM	Account for COVID as an exogenous variable(s)	OK	Irregular	• Do not compare with prior or future periods • OK for relative read across channels/ partners • Corroborate findings with historical data
Sales Lift	Cut short measurement period if needed	OK	Lower/ depressed	• Same as Above
Brand Lift	None	Good	Lower CPPI (Cost per Person Impacted)	• Same as Above
Store Visits Lift	No adjustment possible	OK	C&G foot traffic significantly lower due to lockdowns	• Disregard results for C&G
DAR/TAR	None	Good	Lower cost per unique reach or higher frequency	• Use as-is
Datorama Reporting	None	Good	Lower CPMs and higher engagement	• Use as-is

DETAILED ANALYSIS: MMM

What is MMM?

- Media Mix Modeling (MMM): Attribution method which determines the impact of advertising through statistical correlation with offline sales
- MMM can account for other factors such as price, distribution, media, In-store and external factors (Seasonality, Weather, Competition, Holidays etc.)
- MMM is measured for a single channel at a time and the view is high-level (decompositions are limited)

How is MMM likely to be affected by COVID?

- Advertising performance will likely show **significant deviations** from prior periods due to underlying **changes in consumer behavior**:

Key Behavioral Changes	Impact on Advertising Effectiveness for MMM
Panic buying/ ineffective media	Consumers are buying irrespective of media exposure, reducing the effectiveness of advertising on sales
Increased online buying	Consumers who are influenced by advertising may purchase online, reducing the observed effectiveness of advertising on offline channels
Buying multi-packs and large format stores	Increase in sale of multi-packs and sale at large format stores will negatively impact ROI, as profit is less than single-serve and C&G outlets
Change in consumption occasions	With the absence of events and socializing, volume of sales may be reduced
Increased platform usage	Lower CPMs due to increased platform usage and reduced advertiser demand will positively impact ROIs.
Stock Outs	Retailers may have been “cleaned out” of goods, reducing the effectiveness of advertising on offline sales
Differing consumer behavior by region	Consumer behavior differs by region depending on when COVID peaked which may cause irregularities in results.

- Without any adjustment, MMM models **will not be accurate**. COVID-driven activity may be (in part) incorrectly attributed to an existing driver. However the remedy is fairly straightforward (see below).

DETAILED ANALYSIS: MMM, cont.

How should we adjust our approach?

- In terms of increasing model **accuracy**, external events such as the COVID crisis can be factored into the model as additional explanatory variables; this should help account for the impact of the pandemic
- The PepsiCo ROI Engine Team are aligned on this approach and have incorporated the appropriate variables into their models

What is the verdict on MMM?

- If additional variables are factored into the model to account for COVID, **MMM will be accurate**
- In terms of actual advertising **performance,** results may still be unusual/ extreme due to nontypical consumer behavior
- Therefore, comparisons to **prior and future periods** should be disregarded; **Cross-Channel** comparisons may still be valuable, as they provide a read on relative performance and on the resiliency of each media platform.

DETAILED ANALYSIS: SALES LIFT

What is Sales Lift?

- Sales Lift estimates the impact of marketing activity by comparing sales of ad-exposed HHs ("test") to sales of ad-unexposed HHs ("control"), holding all other factors constant
- Measures offline sales only, limited to one partner and (usually) one campaign at a time

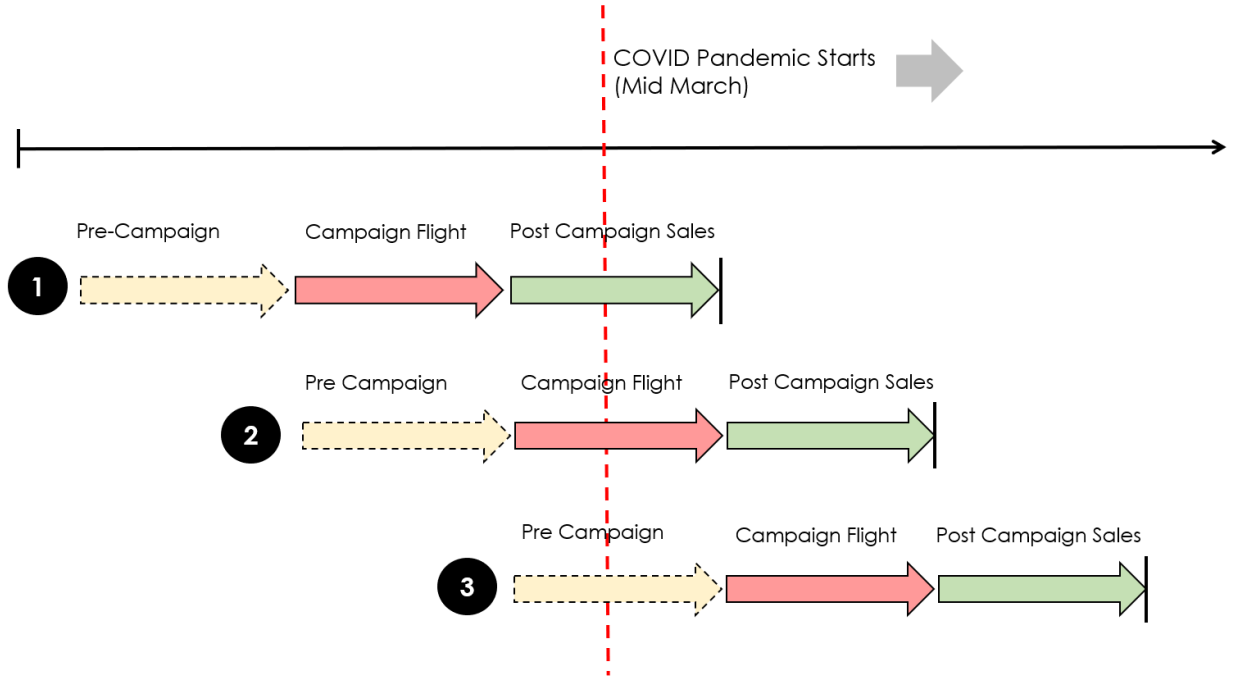
How is Sales Lift likely to be affected by COVID?

- In terms of **accuracy**, Sales Lift is generally **more resilient** than historical comparisons since both the test and control populations are exposed to the same overall conditions
- However, for Sales Lift to be accurate, ad-exposed and unexposed households must be **identically composed** in terms of purchase behavior, demographics, purchase cycle etc. to isolate the impact of advertising from other factors (see below for proposed solutions)
- In terms of **advertising performance**, most of the changes to consumer behavior affecting MMM will also affect Sales Lift (panic buying, online sales, multi-packs, fewer social events, increase platform usage, stock outs, and regional differences)

How should we adjust our approach?

- **Accuracy** for Sales Lift relies on ad-exposed and unexposed households to be **identically composed**
- If ad-exposed and ad-unexposed cells are behaviorally matched leveraging data from pre-COVID periods and are used for measurement during COVID periods, **the cells may no longer be comparable**
- Results should be adjusted based on campaign flight period relative to COVID:

DETAILED ANALYSIS: SALES LIFT, cont.



1. Post campaign during COVID

- Study may not be accurate as test and control cells were matched using data prior to COVID
- Post period sales should be **removed** for both groups; sales performance may be understated as post period sales are not included

2. Pre- campaign before COVID and flight/ post period during COVID

- Study may not be accurate as test and control cells were matched using data prior to COVID and study is run during COVID
- Study results during COVID **should be cut short**; sales performance may be understated as part of the campaign + post period sales are not included

3. All three periods during COVID

- Study will be accurate; pre-campaign HH matching data should account for COVID panic buying behavior.

DETAILED ANALYSIS: SALES LIFT, cont.

Adjustments made by measurement partners:

NCS will use extreme buying, basket size and out of stock variables as match criteria for exposed and control HHs; this will ensure they react similarly during the campaign and post period

IRI will provide reporting break outs based on HHs based on level of panic buying behavior – low, high and no panic buying behavior

ODC will include COVID time period sales metrics at product, brand and category levels with the potential to be used in the process of balancing control and exposed, if they prove to be important

What is the verdict on Sales Lift?

If the appropriate steps are taken with respect to campaign flight periods and control/ exposed matching, results should be **accurate**

In terms of **performance**, results may still be understated as sales periods will be curtailed in some cases (per adjustment above); additionally, changes in underlying behavior may still result in **depressed ad impact**

Therefore, **historical comparisons** won't make sense but can be used to corroborate findings drawn from the COVID period; **Cross-channel** or cross-partner comparisons may still be valuable, as they provide a clean read on relative performance

DETAILED ANALYSIS: BRAND LIFT

What is Brand Lift?

- A measurement solution that quantifies **attitudinal changes** as the difference between responses of ad-exposed and ad-unexposed users (test & control cells)
- Cell samples are randomly selected from the target audience and have equal chance of exposure
- Survey-based approach; both groups participate in the same survey while the campaign is live

How is Brand Lift likely to be impacted?

- In terms of **accuracy**, brand lift should not be affected as it employs at test and control methodology (which are balanced through randomization*)
- Though accurately reported, perceptions may be skewed or distorted due to COVID; consumers' awareness and **attention may be muted** due to the pandemic
- In terms of advertising **performance**, Brand Lift cost per lifted user metrics may be positively impacted due to **lower CPMs** and greater time on platform

How should we adjust?

- No adjustment needed

What is the verdict for Brand Lift?

- Brand lift should be **accurate** in terms of measuring sentiment and perception shift
- However, **advertising performance** may be irregular due to changes in consumer receptivity to messaging (lack of attention, increased ad competition due to more online viewing); also, we may see heightened results in cost/per metrics as inventory will be cheaper
- Brand lift can be used **normally**, but be on the lookout for unusual patterns due to COVID; **cross-channel** and cross-platform should be valid from a relative standpoint; use **historical results** to corroborate findings

* If cells are selected based on (or balanced by) historical data, the effects will be similar to those described in the Sales Lift section

DETAILED ANALYSIS: OTHER MEASUREMENT TOOLS

Foot Traffic Analysis:

- Tracks physical movements to retail outlets; commonly used for campaigns driving to C&G
- Similar to Brand lift in methodology (test/ control)
- Results will be **accurate**, but performance may be **significantly reduced** as consumers are making significantly fewer trips to C&G (and outlets are on lockdown)
- No adjustment is possible; results are likely to be highly skewed and should be **disregarded**

DAR/ TAR:

- Determines overlap and frequency among different advertising channels, including walled gardens and linear TV
- Results will be **accurate**; overlap and frequency may be **greater than usual** due to increased hours spent online among users
- Can be used **without adjustment**

Datorama

- Dashboarding solution which returns digital metrics such as CPM, delivery, video completion, viewability etc.
- Results will be **accurate**; CPMs may be lower due to reduced demand and greater inventory availability during COVID
- Can be used **without adjustment**

CONCLUSIONS

- With suggested modifications, most methods show good **accuracy**—they faithfully reflect how advertising is performing in the marketplace
- However, advertising **performance** may be unusual/ depressed due to changes in consumer behavior and should be viewed with caution
- Therefore, 2020 results will likely **not be directly comparable** to past periods and **not projectable** moving forward; channel results should be comparable in a **relative sense,** but takeaways should be corroborated with **historical findings**

APPENDIX: SALES LIFT STUDIES IMPACTED

Pre-Campaign Period (1 year)	Campaign Flight	Post-Campaign Period (4 weeks)	Impact on Sales Lift
Pre-COVID	Pre-COVID	COVID	G Zero
Pre-COVID	COVID	COVID	KeVita Q1, TPP Q1, IQO Q1, Hots Q1, Chewy Q1, Naked Q1, Cheetos Super Bowl, Doritos Flamin Hot Limon Q1, Doritos Cooler Ranch Super Bowl , Lay's Q1, Lay's Lightly Salted Q1, Lay's Poppables Q1, LKC Q1, Ruffles Lime & Jalapeno Q1, Smartfood Q1, Stacy's Q1, SunChips Q1, Tostitos Q1, Tostitos Innovation Q1, Variety Packs Q1
COVID	COVID	COVID	Efficacy, Quaker Hots Anthem, Quaker Hots Recipe, Mountain Dew's Masterbrand

APPENDIX: SALES LIFT COVID MEASURES

NCS:

Campaign Measurement Enhancements starting Apr 20, 2020 – Adding the following control variables

Basket Size: In addition to matching HHs in the pre-campaign period, HHs who exhibit an increase in their basket size during the analysis period (campaign flight & post period) are bucketed together. This is not specific to the category or brand.

Stock-Out: Adding control for brand and category stock-outs. Stock-outs being defined as "Trip to store on day in which brand/category was purchased a level significantly below previous average daily dollar sale". Current threshold is 2.5 standard deviations.

Extreme Buying: This is specific to the brand & category. HHs who exhibit an increase in their brand & category purchase during the analysis period are bucked together.

IRI:

Campaign Measurement Enhancements – effective April 1st for projects that are commissioned direct from IRI

Media Market level matching between Test and Control: Given that panic buying is essentially a local phenomenon, ensuring that Test and Control are matched based on the media market helps adapt to local purchase behavior changes and ensure homogeneity of test and control

Identifying and Matching households based on "Panic Buying" behavior : IRI already matches Test and Control at the HH level based on purchase behavior metrics such as Retail Banner, ProScores etc. In addition, we will match households in Test and Control based on no panic buying exhibited, low level of panic buying, and high level of panic buying exhibited.

Purchase breaks based on "Panic Buying" behavior (affected campaigns): For affected campaigns, we will report purchase breaks by panic buying behavior to understand ad effectiveness differences among the three consumer segments identified in 2 above.

APPENDIX: SALES LIFT COVID MEASURES, cont.

ODC/MOAT Outcomes:

Causal measurement approaches based on test/control principles (such as MOAT Outcomes) are going to be more resilient to irregular spending behaviors like those we are seeing now.

In addition to the rigorous QC of MOAT Outcomes modeling that has always been in place, we will add planning steps designed to surface any potential measurement challenges before a campaign begins.

We have created additional balance variables to capture COVID time period sales metrics. We will include these metrics at product, brand and category levels with the potential to be used in the process of balancing control and exposed, if they prove to be important. We will closely monitor the importance of these variables for campaigns that ran during peak COVID period.

The ingoing hypothesis of Oracle Data Science teams is that sales lift estimates will not be biased (i.e. consistently higher or lower) but may be less precise (i.e. come with larger confidence intervals).

It is possible (even likely) that consumer responsiveness to advertising will vary by category and potentially by brand.